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Course Name:5G 6G Communication Systems

Course Code:ECA5601

Ex:10 5G Network Slicing Environment Using MATLAB

Aim:

To analyze the performance of a 5G Network Slicing environment by studying the number of network slices, slice utilization, and end-to-end latency using MATLAB.

Objective – 1: Analyze Channel Capacity (bps/Hz)

Analyze the creation and allocation of multiple network slices in a 5G network using MATLAB. Observe how different slices can be configured to support diverse service requirements and applications.

Objective – 2: Compare Signal-to-Noise Ratio (SNR) (dB)

Compare the utilization of different network slices using MATLAB. Analyze how efficiently network resources are allocated and utilized across multiple slices.

Objective – 3: Analyze Antenna Gain (dB)

Analyze the end-to-end latency of different network slices using MATLAB. Compare the latency values to understand the Quality of Service (QoS) requirements of various 5G applications.

Software Required:

- MATLAB

Theory

Network slicing is a key feature of 5G that enables a single physical network infrastructure to be partitioned into multiple logical networks called slices. Each slice is optimized for a specific service category, such as enhanced Mobile Broadband (eMBB), Ultra-Reliable Low-Latency Communications (URLLC), or massive Machine-Type Communications (mMTC). By dynamically allocating network resources, network slicing improves flexibility, resource utilization, and Quality of Service (QoS) while supporting diverse application requirements.

Objective 1: Matlab Code and Output

```
clc;

clear;

close all;

slices = {'eMBB','URLLC','mMTC','Private','IoT'};

count = [1 1 1 1 1]; % One instance of each network slice

figure

bar(count)

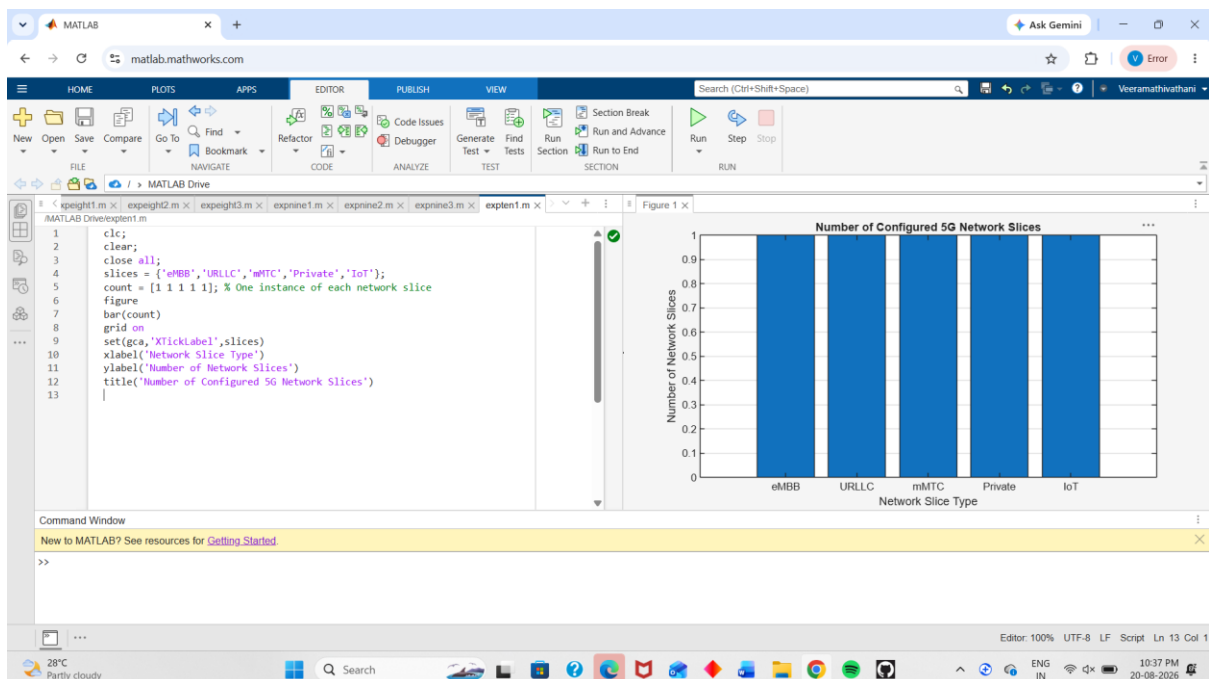
grid on

set(gca,'XTickLabel',slices)

xlabel('Network Slice Type')

ylabel('Number of Network Slices')

title('Number of Configured 5G Network Slices')
```



Objective 2: Matlab Code and Output

```
clc;

clear;

close all;
```

```

slices = categorical({'eMBB','URLLC','mMTC','Private','IoT'});
utilization = [85 70 60 75 50];

figure;
bar(slices, utilization);

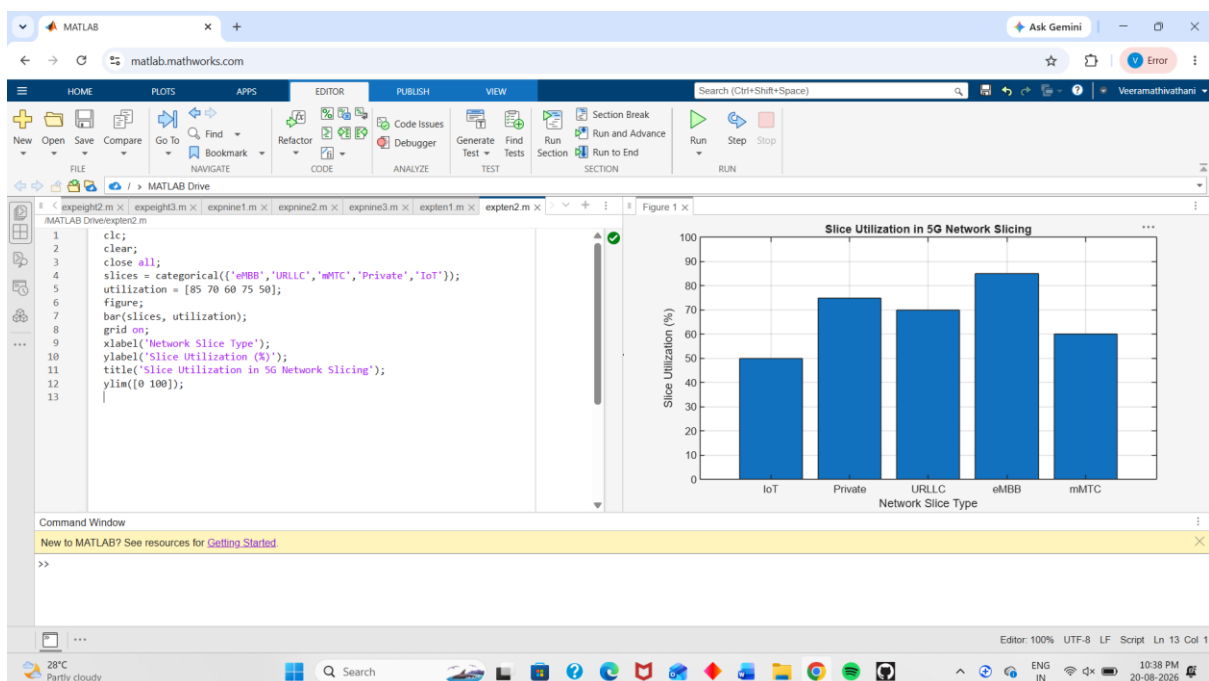
grid on;

xlabel('Network Slice Type');
ylabel('Slice Utilization (%)');

title('Slice Utilization in 5G Network Slicing');

ylim([0 100]);

```



Objective 3: Matlab Code and Output

```

clc;

clear;

close all;

slices = [1 2 3 4 5];

latency = [25 5 40 15 60];

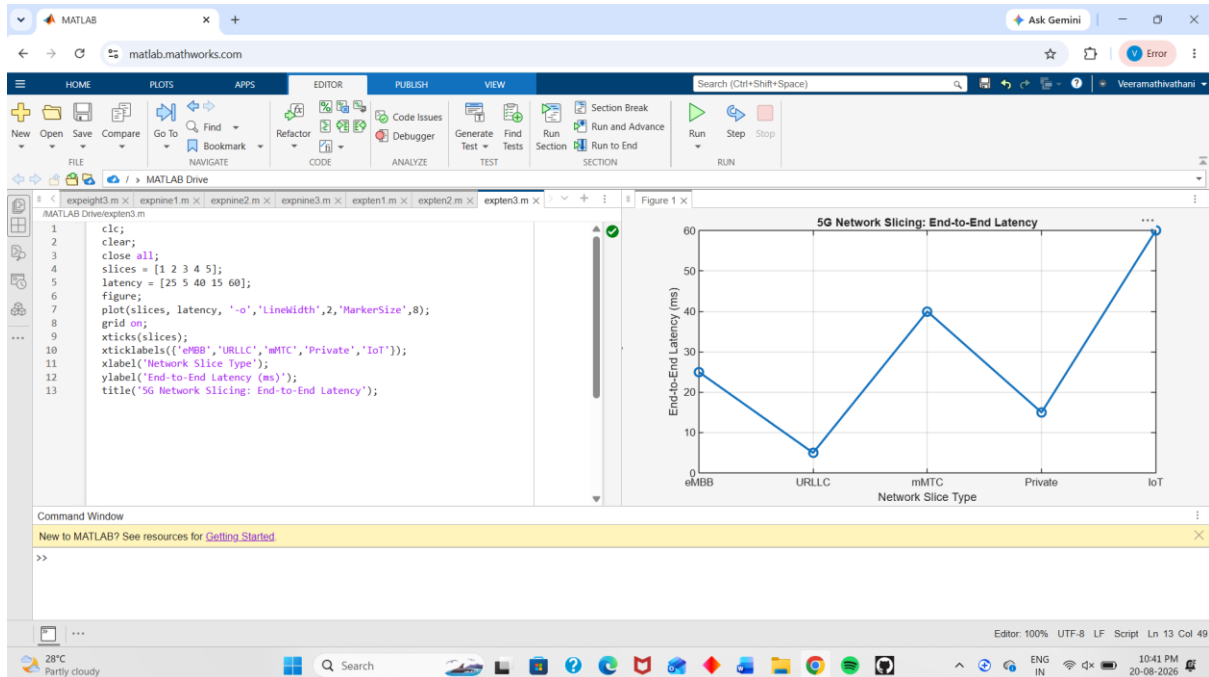
figure;

plot(slices, latency, '-o','LineWidth',2,'MarkerSize',8);

grid on;

```

```
xticks(slices);
xticklabels({'eMBB','URLLC','mMTC','Private','IoT'});
xlabel('Network Slice Type');
ylabel('End-to-End Latency (ms)');
title('5G Network Slicing: End-to-End Latency');
```



Result:

The MATLAB analysis successfully demonstrated the characteristics of a 5G Network Slicing environment. The results showed that multiple network slices can coexist while exhibiting different utilization levels and latency characteristics based on their intended services.